

Minimizing Contractor Reliance & Credit Purchases to Strengthen Direct Customer Relationships and Improve Cashflow in a Plywood and Interior Materials Business

A Mid-Term Report for the BDM Capstone Project

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1. Executive Summary

The organization is a regional distributor of plywood and building materials operating through a mixed B2B and retail model. While contractors were initially believed to be the primary revenue drivers, underlying channel dependencies and operational inefficiencies raised concerns, particularly around sales concentration and sluggish inventory movement.

To evaluate this, data was collected for FY25 (April 2024 to March 2025), including transaction-level sales, purchase records, inventory logs, and contractor performance. Key metadata included columns such as Buyer Role, Purchase Value, Payment Method, Sales Target Hit, Stock Ratio, and Wastage Percentage. Descriptive statistics were computed across formal (registered) vs. informal (unregistered) channels and by buyer type (B2B, Contractor, End Customer), revealing stark differences in average order values, frequency, and variance.

Analytical methods applied included ANOVA for segment profitability, Lorenz Curve and Gini coefficient for contribution concentration, and turnover ratio analysis for inventory performance. Findings revealed that End Customers had the highest per-order profitability. Moreover, just 4% of contractors contributed ~60% of high-tier sales, pointing to fragile channel dynamics. Inventory data also showed inefficient turnover (<60%) and high wastage rates (~14%).

These insights lay the groundwork for deeper diagnostic work in the final phase, with focus areas including contractor diversification, credit risk tagging, and inventory optimization.

2. Proof of Data Originality

All tangible evidence submitted and accessible at:  21f3002068 BDM Project

2.1 Primary Dataset containing Sales, Purchase, Sales/Purchase Invoices, and Contractor Performance:  Primary Raw Data

2.2 Letterhead from SK Timber signed by Mohammad Aslam (Founder):

 Business Letter Head

2.3 Fieldwork Images:  Pictures

2.4 Video Interaction (20 min 53 sec) (Hindi) and Interview Transcript (English):  Video

3. Metadata and Descriptive Statistics

3.1 Metadata

3.1.1 Sales Data (*Apr-24 - Mar-25*)

The sales dataset contains 1941 rows and 7 columns, including variables such as Sale ID, Date, Buyer Name, Buyer Type, Buyer Role, Buyer Description, Payment Method, and Amount.

Metadata explaining key variables:

Key Variable	Data Type	Data Level	Description
Buyer Role	Text	Nominal	Categorizes the customer as Contractor, B2B, or End Customer.
Payment Method	Text	Nominal	Indicates how payment was made—Cash or UPI
Amount (₹)	Float	Ratio	Total value of the sales transaction in rupees.

Table 3.1.1: Sales Data Metadata

3.1.2 Purchase Data (*Apr-24 - Mar-25*)

The purchase dataset includes 829 rows and 5 columns: Purchase ID, Date, Supplier, and Amount Paid.

Metadata explaining key variables:

Key Variable	Data Type	Data Level	Description
Supplier	Text	Nominal	Name of the vendor or entity from which goods were purchased.
Amount (₹)	Float	Ratio	The amount paid to the supplier for the purchase.

Table 3.1.2: Purchase Data Metadata

3.1.3 Contractor Performance Data *(Diwali Released A-List Nov 2023 – Oct 2024)*

The contractor performance dataset (A-List) contains 67 rows with variables Name, Address, Sales Target Hit (₹), and Contractor Commission (₹).

Metadata explaining key variables:

Key Variable	Data Type	Data Level	Description
Sales Target Hit (₹)	Float	Ratio	Value of sales generated through the contractor's influence.
Contractor Commission (₹)	Float	Ratio	The commission amount paid to the contractor by the business.

Table 3.1.5: Contractor Performance Metadata

This A-List contains the **top 11%** of the total contractor base (67 out of 600+ contractors), which are internally categorized into **A, B, and C performance tiers**. Since only the A-List was shared, all performance-based insights (e.g., overdependence on top contributors) are derived solely from this subset and may not represent the full contractor distribution.

The data pertains to the period **November 2023 – October 2024**, which **partially overlaps** with the broader project analysis window of **April 2024 – March 2025**. As such, all interpretations are made with this temporal limitation in mind.

3.1.4 Sales/Purchase Sample Invoices and Inventory *(April 2024 – March 2025)*

The dataset includes transactional records from sales and purchase invoices, totaling approximately 120 entries across three categories: direct customer sales, reseller sales, and purchases. The data variables include Invoice No., Date, Buyer/Supplier Name, Address, Product Description, Quantity, Rate, and Total Amount. Complementing this is the inventory dataset covering the period from April 2024 to March 2025, with monthly records detailing Purchase and Sales quantities and values, Opening and Closing Stock levels, and Holding Costs. Together, these datasets present a comprehensive view of both financial and operational flows in the business.

3.2 Descriptive Statistics

To establish a foundational understanding of the dataset, a range of descriptive statistics was computed across sales, purchase, and performance-related metrics. Key measures included **count, mean, median, minimum, maximum, total**, and **standard deviation**, applied across two dimensions:

- Formal-Informal-level segmentation (Registered vs. Unregistered)
- Buyer role-level segmentation (B2B, Contractor, End Customer)

These statistics provide insights into the distribution, central tendency, spread, and value contribution of transactions within each segment.

Key summary statistics are presented below:

Channel-Level Segmentation			Buyer Role Level Segmentation	
Metric	Registered (All B2B)	Unregistered	Contractor	End Customer
Count	105	1,836	1,336	500
Mean (₹)	₹29,160	₹5,606	₹1,000	₹12,500
Median(₹)	₹30,200	₹1,800	₹1,462	₹16,680
Min(₹)	₹240	₹19	₹19	₹5,010
Max(₹)	₹2,44,213	₹49,982	₹5,000	₹49,982
Sum Total (₹)	₹30,61,811	₹1,02,92,592	₹19,52,632	₹83,39,960
Std. Dev	₹26,733	₹9,078	₹1,320	₹11,381

Table 3.2.1 Channel and Role-Based Sales Distribution Overview

While B2B sales bring in higher value per transaction, the **overall revenue** is still driven primarily by **B2C (Unregistered)** due to the **sheer volume** of transactions.

Higher standard deviation in B2B suggests more volatility and variance in transaction sizes — some purchases are small, some are extremely large. **Lower variance** in B2C means transaction sizes are more **consistent**. The **median gap** is especially telling: typical B2B sales are **~17×** larger than B2C.

Payment Method	Buyer Role	Count	Avg. Amt.(₹)	Median (₹)	Max (₹)	Total (₹)	Share (%)
Cash	Contractor	295	₹678	₹400	₹4,366	₹2,00,084	1.50%
	End Customer	1	₹14,500	₹14,500	₹14,500	₹14,500	0.11%
UPI	B2B	105	₹29,160	₹30,200	₹2,44,213	₹30,61,811	22.93%
	Contractor	1041	₹1,684	₹1,300	₹5,000	₹17,52,548	13.12%
	End Customer	499	₹16,684	₹12,500	₹49,982	₹83,25,460	62.34%

Table 3.2.2 Sales Breakdown by Payment Method and Buyer Role

Most transactions—across all buyer roles—occur via **UPI**. Most contractor payments are UPI, but **low-ticket, high-volume**.

High-value, low-frequency UPI transactions (especially from end customers) may be another key area of credit risk. Without tagging credit explicitly, tracking exposure remains a challenge.

The **net commission** (₹21L) offered to contractors versus the **total sales influenced** (₹1.57Cr) suggests an **overall commission rate of around 13.4%**, which seems reasonable. But given the high spread in Sales Target Hit (₹1.57L), it's likely that **a small fraction of contractors are generating a disproportionately large share of revenue**. This points to dependency risks.

Metric	Contractor Commission	Sales Target Hit
Min	₹2,863	₹90,743
Median	₹24,573	₹1,81,597
Mean	₹31,496	₹2,34,696
Max	₹1,13,897	₹8,49,966

Metric	Average
Sales	₹11,12,867
Purchase	₹10,37,921
Stock Ratio (%)	59.84%
Wastage %	14.06%

Sum	₹21,10,244	₹1,57,24,614
Std Dev	₹21,700	₹1,57,862

Table 3.2.3 [a] & [b]: Descriptive Statistics for Contractor Incentives and Inventory Efficiency

Comparing the average of sales and purchases suggests a relatively low gross profit margin (~6.74%), which may indicate tight margins or high cost of goods sold (COGS).

Stock turnover ratio% tells us that on average, only ~60% of stock is moving, highlighting an inefficiency in inventory utilization. Monthly wastage level (deadstock + damaged) hovered around 14%, reinforcing the need for tighter stock rotation, forecasting, and shelf-life monitoring. Overall, these descriptive insights lay the groundwork for deeper investigation.

4. Detailed Explanation of Analysis Process/Methods

4.1 Data Extraction and Cleaning

4.1.1 Extracting Tabular Data from Scanned Documents

Sales, Purchase, and Contractor Performance data were initially available as photographs taken using a mobile phone. To convert these images into a usable spreadsheet format, a hybrid method was employed. The “Scan Text” feature of the native Photos app in Windows 11 was used to extract textual content from the images.

This text was either fed into a Python script ( Data Extraction Python Script) that automated the structuring into Excel files or, in cases where the data was limited in volume, such as the 16 sales invoices and 11 purchase invoices, it was manually organized into spreadsheets. The latter approach ensured accuracy in data capture and required much less data cleaning than the former.

Once extracted, all the data were ready for subsequent cleaning and preprocessing.

4.1.2 Data Cleaning

Monetary values in the “Amount (₹)” column were rounded to remove decimals for clarity. OCR-related artifacts and duplicates were cleaned using Google Sheets’ find-and-replace (e.g., fixing misread characters in names and labels). Column names were standardized across datasets for clarity—e.g., “Target (₹)” → “Sales Target Hit (₹)”, “CC (₹)” → “Contractor Commission (₹)”, and “Particular” → “Supplier”.

4.2 Preprocessing and Feature Engineering

In the Contractor Performance dataset, entries such as “ZABBAR BHAI 385” were parsed to separate the contractor's name and ID into distinct fields by introducing a dedicated **Contractor ID** column. Similarly, in the Sales dataset, additional variables were engineered by extracting key features such as **Buyer Type** (Registered/Unregistered), **Buyer Role** (B2B, Contractor, End Customer), **Buyer Description**, and **Payment Method** (Cash or UPI) to enable more granular analysis.

4.2.1 Filling Data Gaps with Business Owner Disclosures

To fill critical data gaps, the business owner shared key operational estimates. Around 30–40% of unregistered sales are made on credit. Wastage runs 15–20% monthly (mostly deadstock), and credit-related losses are estimated at 5% annually. The complete contractor performance list accounts for total sales of approximately ₹3.5 crore across all contractors. All registered sales (i.e, B2B) are prepaid. Additionally, the proprietor provided a practical sales-based buyer role classification criterion such that high-value purchases (>₹5,000) are typically made by direct customers, whereas lower-value transactions (≤₹5,000) are associated with contractors.

4.2.2 Customer Segmentation and Channel Interpretation

The initial conversations with the business owner suggested high, large-ticket direct sales to contractors, and the lack of proper classification between end customers and contractors in unregistered sales reinforced this assumption. However, further clarification from the business owner, it was revealed that contractors primarily function as **demand aggregators**

rather than bulk purchasers. The proprietor clarifies, approximately 80% of **end customer** sales are contractor-**influenced**.

Using the classification method in section 4.2.1, the business’s customer base is classified into **three distinct segments (B2B, Contractors, End Customers)** instead of two (B2B and B2C). As shown in Table 4.1 below:

Segment	Description	Credit Usage	Role
End Customers	Homeowners buying directly	30–40% (Unregistered)	Revenue source
Service Providers (Contractors)	Carpenters, interior designers	Low (mostly UPI/Cash)	Demand aggregators, not buyers
Resellers (B2B)	Registered businesses	0% (Full payment upfront)	Buyers, revenue drivers

Table 4.1 Three-Segmented Model

While contractors and resellers are both non-end users, their financial behavior and influence on sales differ significantly. **Contractors act as facilitators; resellers act as independent buyers.** Hence, **they must be treated as separate channels** when analyzing risk, sales efficiency, and channel resilience. This reframes our problem not as margin leakage but as **channel dependence**. Hence, problem statement 1 pivots from pricing power to contractor-channel efficiency and risk exposure.

4.2.3 Estimating Inventory Data

To compensate for the absence of formally maintained inventory records, key inventory metrics were estimated using sales and purchase data, weighted average pricing from sales/purchase invoices, and business-owner insights.

Units were standardized (e.g., 1 PCS = 1 UNIT; 250 RFT = 1 UNIT) to allow unified calculations. Core variables such as monthly sales/purchase units, wastage of 18%, opening stock (Initialized April at 1.2× average sales, buffer stock taken 20% as per industry standard), and closing stock calculated using the inventory balance formula:

$$\text{Closing Stock Unit} = \text{Opening Stock Unit} + \text{Purchase Unit} - \text{Sales Unit} - \text{Wastage Unit}$$

The stock turnover ratio, holding cost (1.67% monthly on ₹400 average selling price), and excess stock (benchmarked against a 3-month moving average) were derived using standard formulas. This structured estimation enabled a foundational inventory reconstruction for further volatility analysis.

4.3 Analysis Process and Methodologies

To address Contractor Channel Fragility, Hidden Credit Risk Exposure, and Seasonal Demand Volatility, it was required to assess whether reliance on contractors is even financially justifiable. If not, then which segment is most profitable? Understand what % of contractors are contributing to the majority of the A-List sales. And assess the seasonality in customer demand.

4.3.1 ANOVA Test for Segment Profitability Analysis

To understand whether reliance on contractors is financially justifiable, a **one-way ANOVA test** was conducted to evaluate if the **Average Order Value (AOV)**—defined as the mean sales amount (₹)—differs significantly across the three buyer segments: **Contractors**, **B2B**, and **End Customers**.

Null Hypothesis (H_0): ($\mu_1 = \mu_2 = \mu_3$) There is no significant difference in AOVs among the three segments.

ANOVA compares the variance between segment means relative to the variance within segments. The F-value and p-value were computed to assess whether the observed differences in average sales values were statistically significant.

The **F-value** quantifies how much the group means differ relative to the variation within each group — a higher F-value generally signals more pronounced differences. The **p-value** indicates the probability that these differences occurred by random chance. A low p-value (below 0.05) would suggest that at least one segment's average order value (AOV) significantly differs from the others.

4.3.2 Contractors' Contribution Analysis and Lorenz Curve

Since only the A-list—the top 11% of contractors (67 individuals)—was available, a **contribution analysis** was conducted to assess internal concentration. Cumulative sales contributions were calculated for the top 5, 10, 20, 50, and all 67 contractors, expressed as percentages of the total A-list sales.

To further quantify the degree of concentration, a **Lorenz curve** was plotted and a **Partial Gini coefficient** computed—"partial" in the sense that it is based only on this subset of the contractor population (i.e., the top 67 out of 600+).

This approach enabled the assessment of potential dependency risks within the Contractor channel, based on the distribution of sales among top-performing contributors.

4.3.3 Monthly Trend Analysis: Inventory, Demand Seasonality, and Margin Signals

To supplement buyer segmentation insights, a month-wise trend analysis was conducted for FY25 (April 2024 to March 2025) to understand how purchase and sales amounts move in tandem—or not—over time. This aimed to identify margin stress periods, inventory planning misalignments, and latent seasonal demand cycles.

4.3.4 Buyer Role Profiling Through Transaction Patterns.

Buyer Role	Count	Average Order Value (₹)	Total Sales (₹)	Share (%)
B2B	105	₹29,160	₹30,61,811	22.93%
Contractor	1336	₹1,462	₹19,52,632	14.62%
End Customer	500	₹16,680	₹83,39,960	62.45%
Total	1941	₹6,880	₹1,33,54,403	100%

Table 4.3.3: Buyer Role–Wise Sales Distribution and Profitability Summary

While end customers account for most of the revenue (62.45%), contractors dominate transaction **volume** (~70% transactions), primarily through **low-ticket, high-frequency**

sales. B2B buyers have the highest average spend (₹29,160) but contribute less to total revenue due to low volume.

5. Results and Findings

5.1 The ANOVA test conducted for Segment Profitability Analysis (available at:  SalesData (Apr-24 - Mar-25).xlsx) revealed a **statistically significant difference** in average order values (AOV) across buyer segments ($F = 105.867 > F\text{-critical} = 2.997$, $p \approx 0$).

- End Customers showed the highest profitability (~₹4,297),
- B2B buyers were moderate (~₹1,577),
- Contractors had the lowest (~₹1,006).

Greater variance in B2B and End Customer segments suggests more fluctuation in order values compared to Contractors. This is evident in the Buyer Segment Sales Trend Graph given below, along with the ANOVA results:

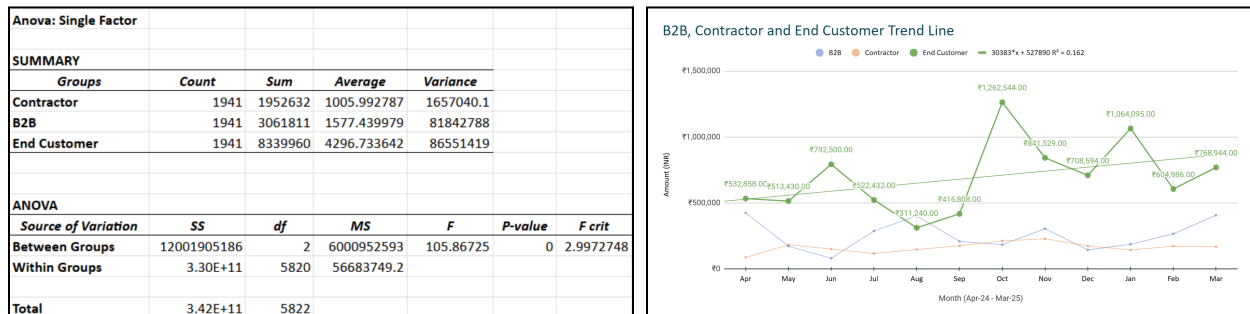


Figure 5.1[a] & [b]: ANOVA Results and Line Chart displaying End-Customer Seasonal Volatile Demand. Much less volatility is observed in the contractor segment, compared to end-customer and B2B sales across the year.

5.2 The contribution analysis revealed a concentration of sales within the Contractor channel:

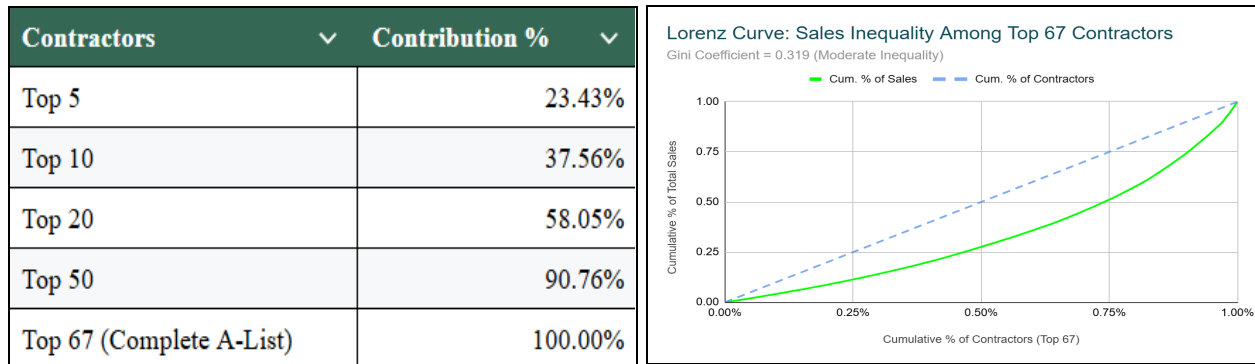


Figure 5.2[a] & [b]: Contribution Breakdown of Top 67 Contractors and Lorenz Curve (Gini=0.329)

Notably, just 20 contractors, less than 4% of the original pool, generated ~60% of the **A-list revenue**. This steep distribution was further validated by the Lorenz curve, which deviated substantially from the line of equality, and a **Partial** Gini coefficient of 0.316, indicating moderate inequality in sales contribution even within the already elite subset. Together, these results underscore a potentially fragile channel structure, heavily reliant on a small core of high-performing contractors. ([+ Contractor Performance Data](#)).

5.3 The inventory dataset shows significant variation in purchase and sales patterns across months. Some key trends:

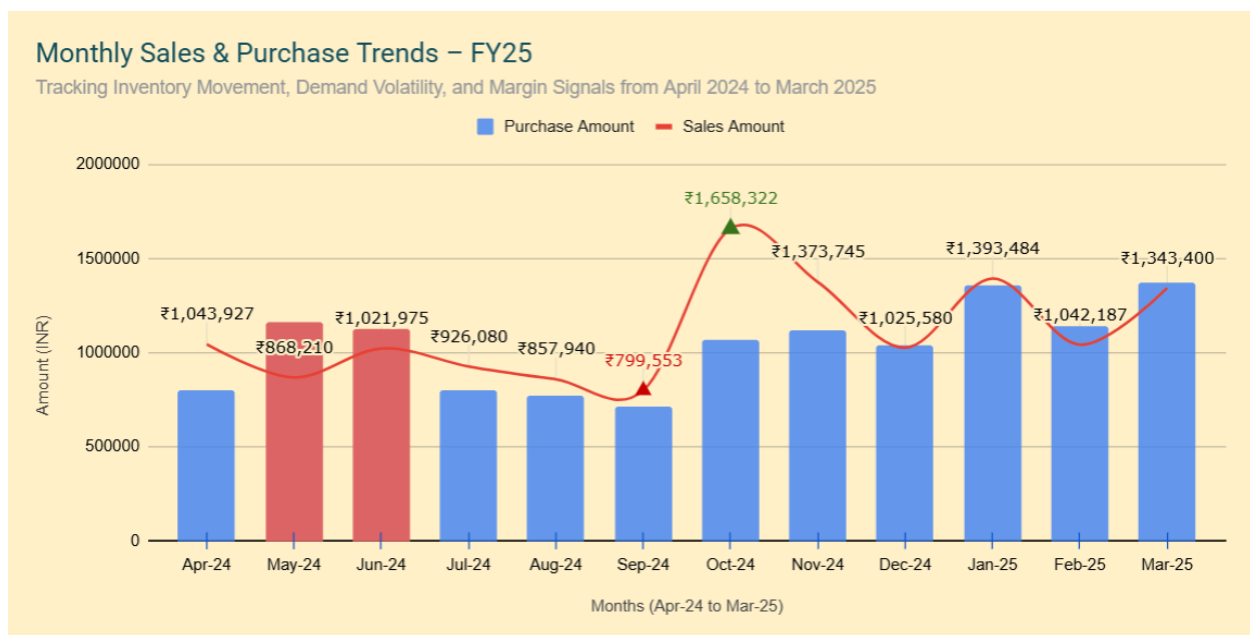


Figure 5.3: FY25 Inventory Dynamics: Sales vs. Purchase Trends (Apr 2024 – Mar 2025)

The graph highlights a significant drop in sales from April (₹1,043,927) to September (₹799,553), a monsoon dip followed by a sharp rebound in October (₹1,658,322). Interestingly, while purchase volumes (blue bars) remained relatively stable and gradually increased, sales (red line) demonstrated far more volatility. The gap between purchase and sales amounts in months like September and October suggests inventory buildup followed by sudden demand spikes, potentially festive-season driven.

Months like May and June (red bars) were purchase-heavy with low sales, leading to stock buildup (e.g., 48.62% - 46.70% turnover). October saw better alignment (80.58% turnover)

 **Inventory Data (Apr-24 - Mar-25) (Estimated)** . Ratios below 50% in several months indicate **inefficient stock movement**. This is especially problematic when paired with consistent **double-digit wastage rates**, primarily due to **deadstock and damage**. Monthly Holding Cost calculations, based on a standard 1.67% monthly holding cost rate and ₹400/unit weighted selling price calculated using sales invoices, reflect a significant inventory carrying burden.

5.4 The business is currently **propped up by a small base of high-value end customers**, while **contractor-driven volume keeps the engine running**, but with diminishing returns. There's a **fragile dependency** on contractors that may not be financially justified at scale. B2B, while small, holds **untapped potential** if nurtured correctly.